

Modern Cloud-Native Data Science

Exploring innovative data science using cloud-native technologies



Francesco Tonini, PhD, GISP

Geospatial Data Scientist

IT Conservation & Geospatial Systems

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1. Pillars of Cloud-Native Data Science

Elastic Compute and Storage

Cloud-native platforms provide **elastic compute** and **flexible storage**, overcoming local hardware limits for **scalable** data science.

Elastic = automatic and dynamic adjustment of computing resources to match demand up or down

Integrated Ecosystem for AI/ML

Unified environments consolidate **notebooks**, **pipelines**, and **governance**, simplifying the entire **spatial modeling** and **machine learning** lifecycle.

Support for Complex Workloads

Cloud-native data science supports complex tasks like **distributed computing** and **real-time inference at scale** for ML workflows.

Democratizing Enterprise Compute

Cloud-native methods **democratize access** to enterprise-grade compute, enabling **faster innovation** and **reducing bottlenecks**



2. Traditional vs. Cloud-Native Approaches: On-prem systems

Hardware and Challenges

Traditional setups depended on costly hardware and slow scaling processes that hindered rapid expansion.

Limited Collaboration

Teams shared local scripts and files manually, resulting in inefficient collaboration and version control

Reproducibility and Deployment Issues

Different environments caused dependency conflicts and complicated model deployment in production

Limited Advanced Experimentation

Large datasets and distributed computing needs were hard to meet, limiting innovation and experimentation

2. Traditional vs. Cloud-Native Approaches: Cloud-native systems

Elastic Scalability - Dynamic Compute Scaling

Elasticity allows instant scaling of compute and storage to meet dynamic workload demands efficiently

Elastic Storage Solutions

Object storage solutions provide cost-effective, petabyte-scale data storage with flexible scalability

Accelerated Data Science Workflows

Elasticity empowers data scientists to process large datasets and run multiple experiments without infrastructure limits

Collaborative Cloud Workspaces - Reproducibility

Cloud-based workspaces enable seamless sharing of notebooks, datasets, and models to enhance team collaboration

3. Operational Excellence & Governance

Cloud Security Features

Cloud platforms offer identity management, encryption, audit logs, and compliance certifications to ensure data protection

Version Control Benefits

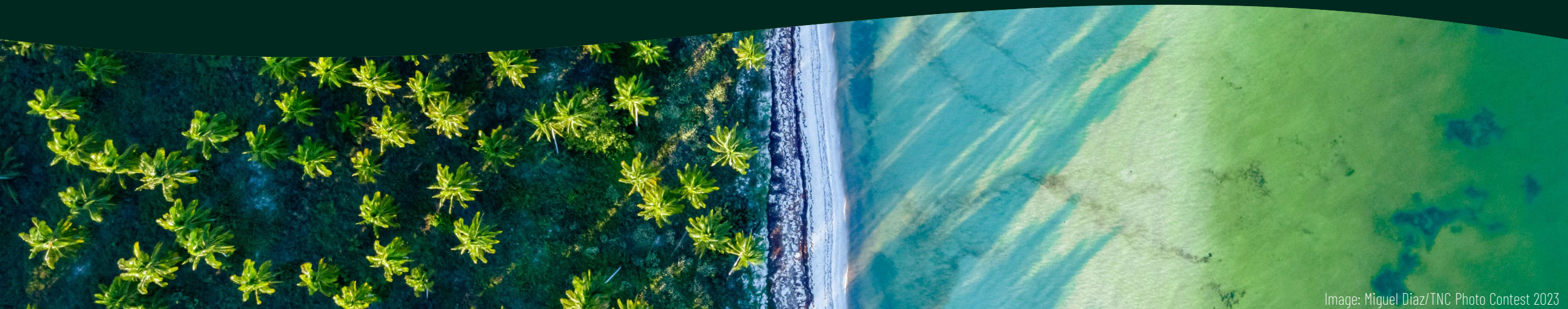
Versioning tools track changes in datasets, models, and pipelines, enabling lineage and rollback

Reproducibility & Reliability

Reproducibility supports reliability, transparency, and regulatory compliance in scalable systems

Governance Frameworks

Governance frameworks provide oversight of datasets, models, and pipelines to reduce risks and promote responsible use of models and AI/ML tools



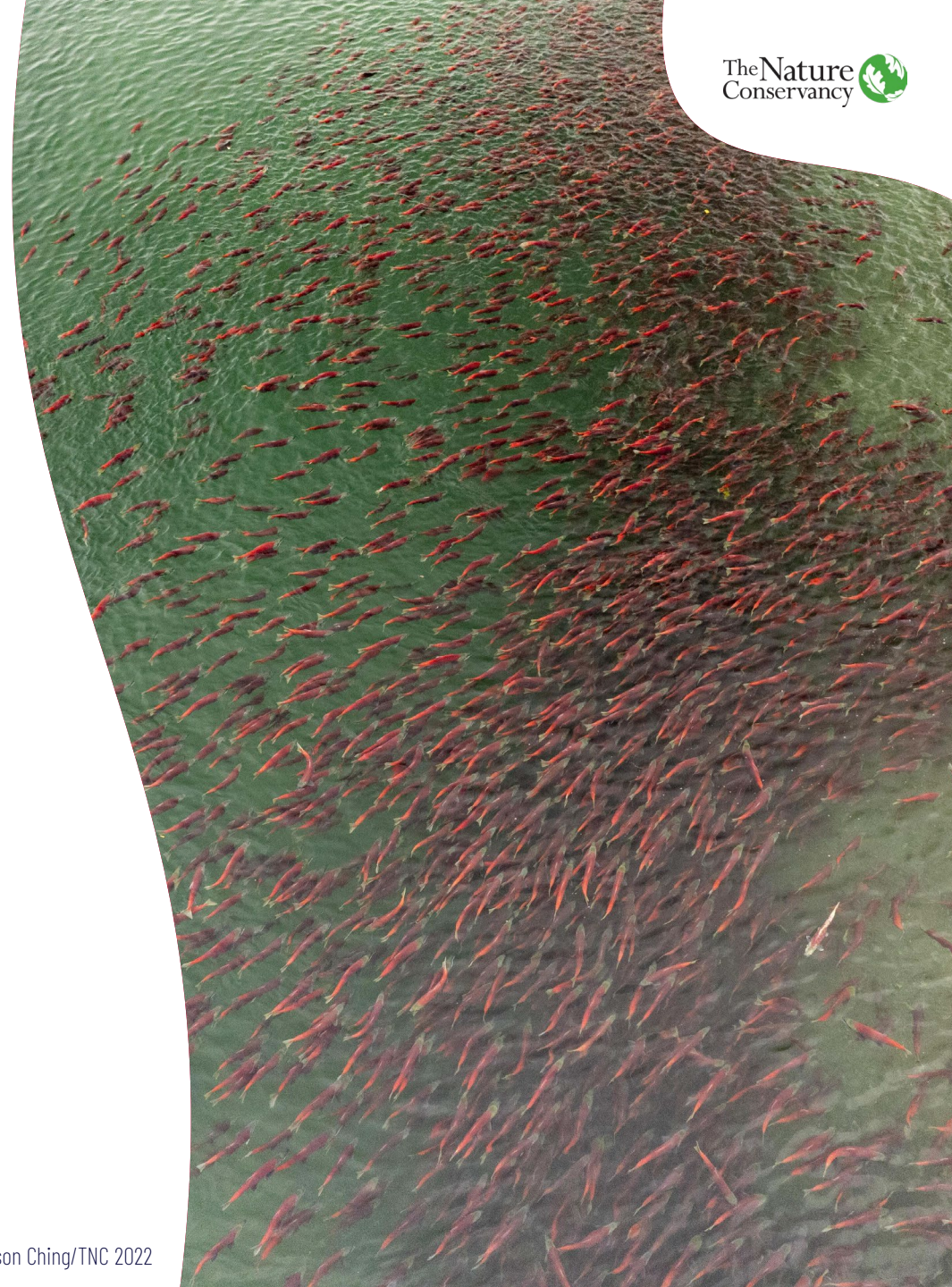
4. Cloud-Native Formats & Protocols

Parquet

Apache Parquet is an open-source, **columnar storage format** optimized for fast data processing and compression, commonly used in big data ecosystems. GeoParquet is an extension of Parquet that **standardizes the storage of vector geospatial data** (points, lines, polygons) within this format. It adds crucial spatial metadata and **supports efficient spatial queries** (e.g., within bounding boxes) for cloud-native applications.

Why?

- **Data compression:** by applying various encoding and compression algorithms, Parquet file provides reduced memory consumption
- **Columnar storage:** paramount importance in analytic workloads, where fast data read operation is the key requirement
- **Language agnostic:** developers may use different programming languages to manipulate the data in the Parquet file
- **Open-source format:** meaning, you are not locked with a specific vendor
- **Legacy Compatibility:** works well with traditional GIS software



4. Cloud-Native Formats & Protocols

Cloud Optimized GeoTIFF (COG)

A specialized GeoTIFF raster file designed for **efficient, fast access** in cloud environments.

Why Cloud Optimized GeoTIFF?

- **Efficient Imagery Data Access:** COG-aware software can stream just the portion of data that it needs, improving processing times and creating real-time workflows previously not possible
- **Reduced Duplication of Data:** Accessing COGs with cloud workflows enables diverse software to all access a single file online instead of needing to copy and cache the data
- **Legacy Compatibility:** Traditional GIS software is able to treat Cloud Optimized GeoTIFFs just like normal GeoTIFFs, so data providers need only produce one format

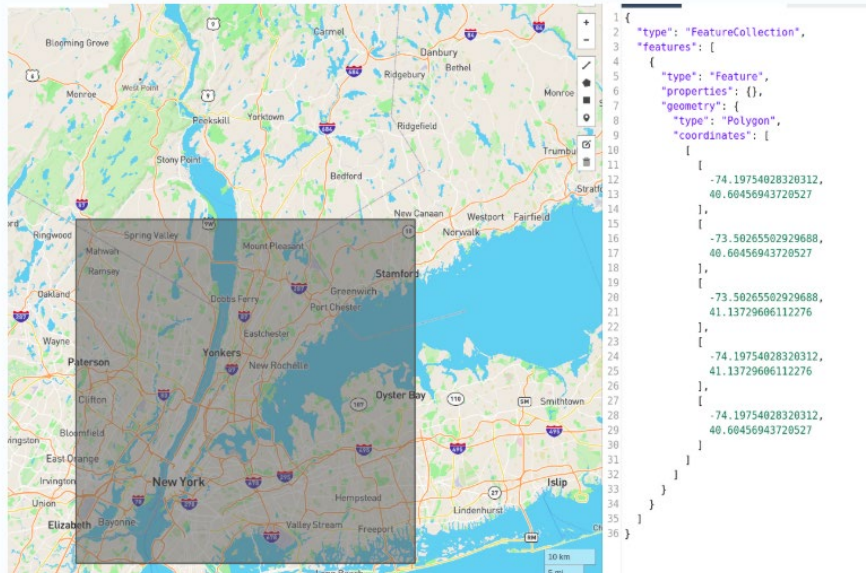
4. Cloud-Native Formats & Protocols

SpatioTemporal Asset Catalog (STAC)

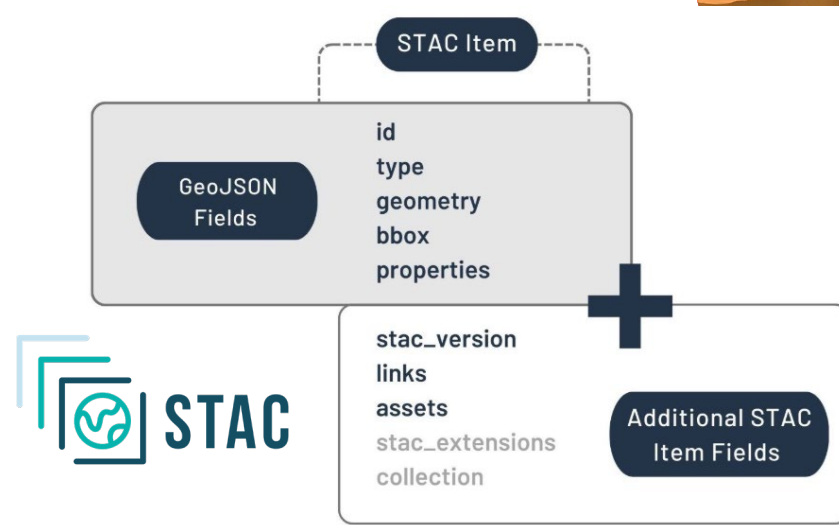
Open specification that **standardizes** how **geospatial metadata** is structured, indexed, and queried.

Why STAC?

STAC provides a common JSON-based language to describe satellite imagery, aerial photos, and other data, allowing users to **efficiently search, browse, and access massive datasets across different platforms**.



Example of an image footprint (left) and its associated GeoJSON file (right).



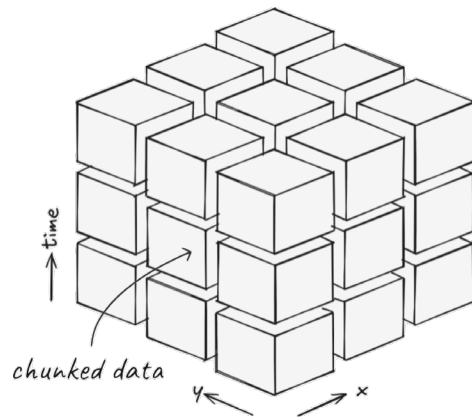
4. Cloud-Native Formats & Protocols

Zarr

Open-source format and specification for storing large, chunked, N-dimensional arrays

Why Zarr?

It was designed for cloud-native, parallel computing, allowing efficient storage and access in memory, or in object stores like S3. It functions like a cloud-ready, chunked, compressed array, breaking data into smaller pieces (chunks) for fast, concurrent I/O, making it ideal for big data science, climate modeling, and AI



```
Metadata: {  
  "spatial_extent": [10.0, 20.0, 30.0, 40.0]  
  "version": 2.0  
}
```



Zarr

4. Cloud-Native Formats & Protocols

Icechunk

Open-source, cloud-native transactional storage engine designed to enhance [Zarr](#) for managing large-scale, multi-dimensional tensor data (e.g., climate, AI, geospatial)

Tensor = fundamental data structure in computing and machine learning that is best understood as a **multidimensional array**

Why Icechunk?

Developed by [Earthmover](#), it provides Git-like version control, branching, atomic updates, and high-performance access on cloud storage like S3



5. Advanced Cloud-Native Data Science Workflows

- **Near-real time deforestation risk & alerts**

Goal: flag emerging forest loss, triage patrols, trigger notifications

Ingest: imagery (satellite/aerial/UAV) scenes delivered to an AWS S3 “hot” storage bucket or access via open STAC catalogs and serve as COGs

Process: analyze with batch/notebook jobs (Python/R/Spark) on AWS SageMaker, train deep learning models, publish and host daily output as a web service in ArcGIS Online

Alerts: serverless (AWS Lambda) when risk crossed AOI threshold [*serverless = cloud-native development model where developers build and run applications without managing underlying infrastructure*]



Image: Nguyen Minh Quang/TNC Photo Contest 2022

5. Advanced Cloud-Native Data Science Workflows

- **Camera trap detection at scale**

Goal: collapse months of manual review into hours, enable “minutes-to-insight” for invasive detections

Ingest: photo and video streams directly into an AWS S3 “hot” data bucket along with metadata

Process: trigger serverless pipelines (AWS Lambda) for detection and inference

Human-in-the-loop: review UI, curate species events, archive imagery

Product: species presence/activity layers and APIs for downstream analysis



5. Advanced Cloud-Native Data Science Workflows

- **Edge-to-cloud bioacoustics for biodiversity monitoring**

Goal: assess species presence and seasonal activity trends with rapid alerts

Ingest: collect audio, stream or batch upload recordings to an AWS S3 “hot” storage bucket

Process: classify with managed inference endpoints (e.g. AWS SageMaker), store detection events as Parquet data files in a data lakehouse

Alerts: notify decision-makers on target species detections, publish time-series heatmaps to ArcGIS Online



5. Advanced Cloud-Native Data Science Workflows

- **“Dark vessel” detection in Marine Protected Areas (MPAs)**

Goal: detect non-AIS (Automatic Identification System) vessels encroaching MPAs and support enforcement

Ingest: SAR (Synthetic Aperture Radar) imagery and optical imagery, joined with existing AIS location data

Process: run object-detection ML pipelines on AWS SageMaker, fuse with AIS location to classify “dark” vessels and produce incident locations

Product: near-real-time dashboards and weekly incident summaries

Image: Kiliyi Yuyan / TNC



6. Future Trends

AI Agents for Workflow Automation

AI agents autonomously manage workflows, optimize resources, and enhance decision-making, minimizing manual effort

Generative AI in Data Science

Generative AI automates code generation, feature engineering, and model tuning to transform data analysis

Serverless and Event-Driven Architectures

Serverless and event-driven pipelines simplify infrastructure management for scalable real-time analytics

Serverless = cloud-native development model where developers build and run applications without managing underlying infrastructure (note: it still relies on servers somewhere!)

Hybrid Cloud Governance

Hybrid and multi-cloud strategies require strong interoperability and governance for secure data operations

6. Key Takeaways

Transformative Cloud Approach

Cloud-native data science overcomes legacy system limits and enables **scalable, collaborative innovation**

Strategic Business Imperative

Adopting cloud-native principles is essential for competitiveness in an AI-driven digital business landscape

Driving Smarter Decisions

Leveraging **modern data stacks** empowers organizations to harness data for **sustainable growth and smarter decisions**